

FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering

Assignment 01

" Git, Gitea, GitHub, Git LFS, and GitHub Pages "

Name	**:**	**Hadia Eman Malik**
Registration No.	**:**	**FA24B1-SE-025**
Course	**:**	**Cloud Computing**
Course Code	**:**	**BSE 665**
Section	**:**	**A**
GitHub Username	**:**	**hadiaemaanmalik**

This report documents the practical work for Assignment 01. All terminal work for Tasks 1, 2 and 3 was done on the Ubuntu server, and the terminal prompt shows the username hadiaemaanmalik and the hostname ubuntu.

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea on the Ubuntu Server

- **Verify the installations**

After reconnecting, Git, Docker and Docker Compose all display their installed versions.

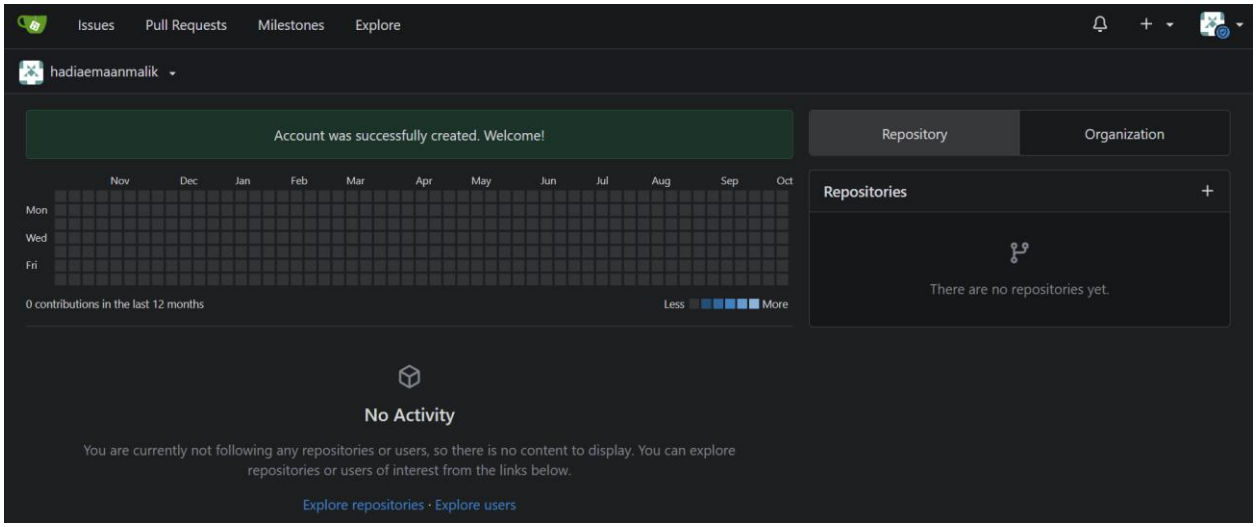
```
Last login: Sat Oct 10 13:48:54 2026 from 192.168.192.1
hadiaemaanmalik@ubuntu:~$ git --version
git version 2.53.0
hadiaemaanmalik@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
hadiaemaanmalik@ubuntu:~$ docker compose version
Docker Compose version v5.6.0
hadiaemaanmalik@ubuntu:~$
```

```
hadiaemaanmalik@ubuntu:~$ git --version
git version 2.53.0
hadiaemaanmalik@ubuntu:~$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
```

- **Clone the Gitea setup repository and start Gitea**

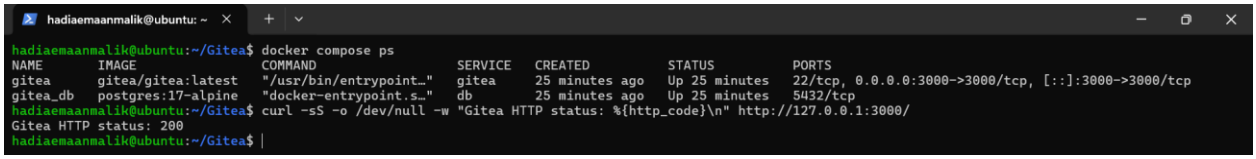
```
hadiaemaanmalik@ubuntu:~$ cd Gitea
hadiaemaanmalik@ubuntu:~/Gitea$ docker compose up -d
[+] up 24/25
 ✔Image gitea/gitea:latest Pulled
 ✔Image postgres:17-alpine Pulled
 ✔Network webnet           Created
 ✔Volume gitea             Created
 ✔Volume gitea_postgres    Created
 ✔Container gitea          Started
 ✔Container gitea_db       Started
hadiaemaanmalik@ubuntu:~/Gitea$
```

- **Open Gitea and create an account**



- **Confirm that both containers are running**

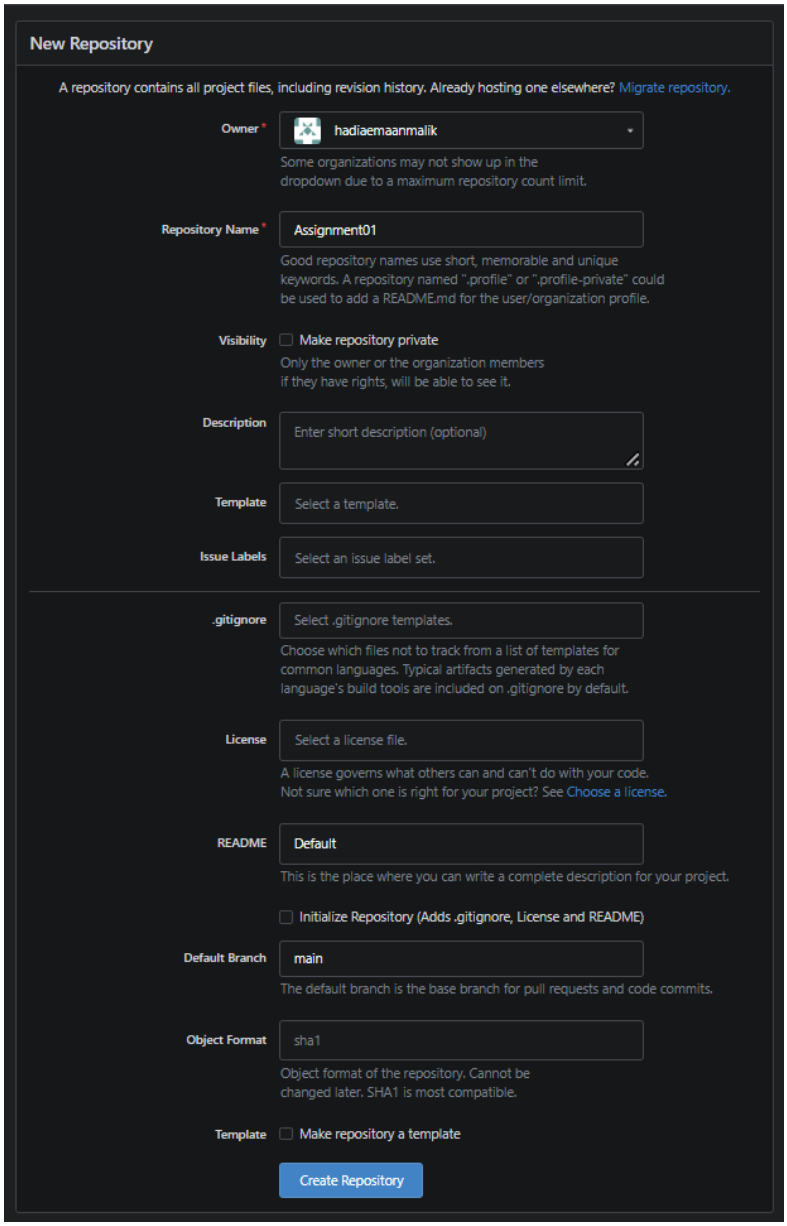
Both the Gitea and PostgreSQL containers are running, and Gitea responds locally on port 3000 with HTTP status 200.



Step 1.2: Create and Push the Initial Repository

- **Create an empty public repository in Gitea**

An empty public repository named Assignment01 was created in Gitea without a README, .gitignore or license.



- **Create the local repository on the Ubuntu server**

```
hadiaemaanmalik@ubuntu:~/Gitea$ cd ~
hadiaemaanmalik@ubuntu:~$ mkdir Assignment01
hadiaemaanmalik@ubuntu:~$ cd Assignment01
hadiaemaanmalik@ubuntu:~/Assignment01$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/hadiaemaanmalik/Assignment01/.git/
hadiaemaanmalik@ubuntu:~/Assignment01$ git branch -M main
hadiaemaanmalik@ubuntu:~/Assignment01$
```

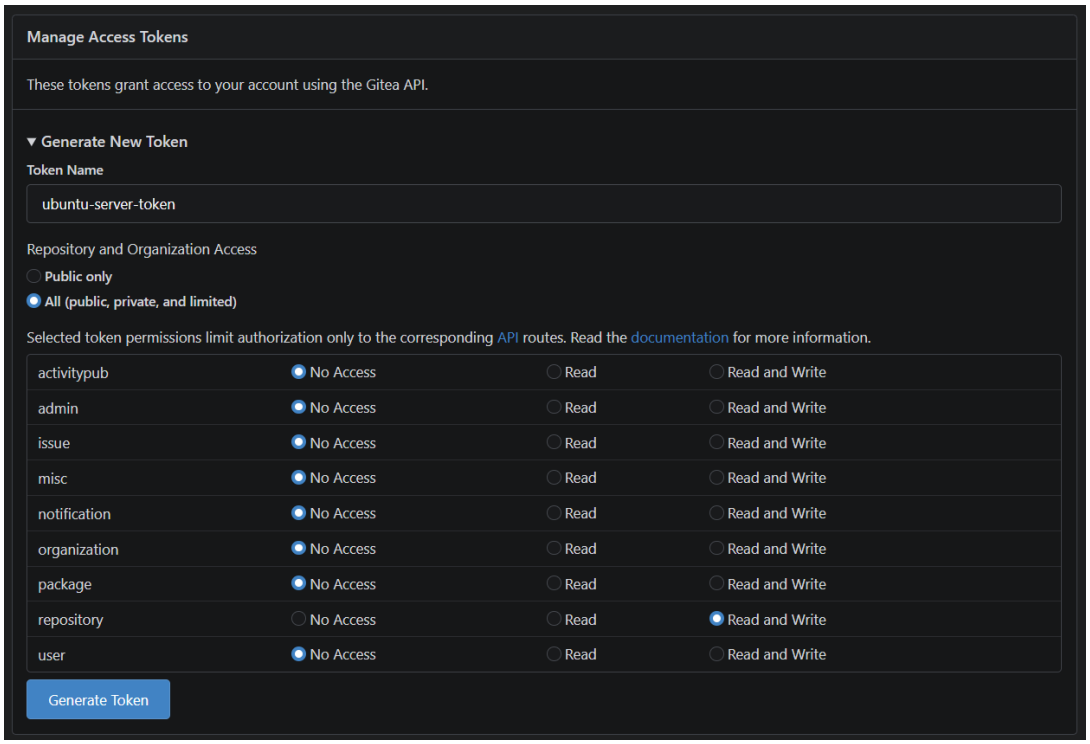
- **Create and commit README.md**

```
hadiaemaanmalik@ubuntu:~$ cd ~/Assignment01
hadiaemaanmalik@ubuntu:~/Assignment01$ git config --global user.name
Hadia Eman Malik
hadiaemaanmalik@ubuntu:~/Assignment01$ cd ~/Assignment01
hadiaemaanmalik@ubuntu:~/Assignment01$ printf "Full Name: Hadia Eman Malik\nRegistration Number: FA24B1-SE-025\n" > README.md
hadiaemaanmalik@ubuntu:~/Assignment01$ cat README.md
Full Name: Hadia Eman Malik
Registration Number: FA24B1-SE-025
hadiaemaanmalik@ubuntu:~/Assignment01$
```

```
hadiaemaanmalik@ubuntu:~/Assignment01$ git add README.md
hadiaemaanmalik@ubuntu:~/Assignment01$ git commit -m "Add student information"
[main (root-commit) 902589e] Add student information
 1 file changed, 2 insertions(+)
 create mode 100644 README.md
hadiaemaanmalik@ubuntu:~/Assignment01$
```

- **Generate a Gitea personal access token**

A personal access token named ubuntu-server-token was generated in Gitea (Settings, Applications, Manage Access Tokens) with Read and Write repository permission. The token is not shown here and was not placed in any command or remote URL.



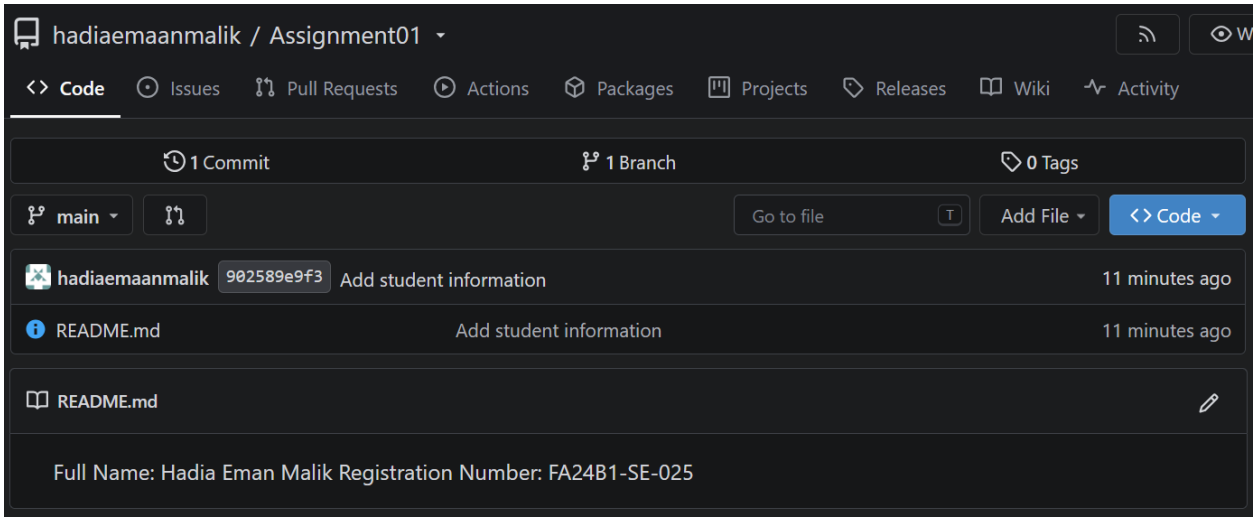
- **Add the gitea remote and push**

The gitea remote was added with my username but without the token. The remote URL contains no password or token. The push succeeded, and the token was entered only at the password prompt.

```
hadiaemaanmalik@ubuntu:~/Assignment01$ git remote add gitea http://hadiaemaanmalik@127.0.0.1:3000/hadiaemaanmalik/Assignment01.git
hadiaemaanmalik@ubuntu:~/Assignment01$ git remote -v
gitea   http://hadiaemaanmalik@127.0.0.1:3000/hadiaemaanmalik/Assignment01.git (fetch)
gitea   http://hadiaemaanmalik@127.0.0.1:3000/hadiaemaanmalik/Assignment01.git (push)
hadiaemaanmalik@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://hadiaemaanmalik@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 293 bytes | 32.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/hadiaemaanmalik/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
hadiaemaanmalik@ubuntu:~/Assignment01$
```

Step 1.3: Verify the Gitea Repository

The Assignment01 repository in Gitea is public and shows the repository owner and name. The rendered README.md shows my full name and registration number, and the Gitea URL is visible in the browser.



Task 1 result: Gitea was installed and run on the Ubuntu server, and a separate local repository was pushed to it over HTTP using the server's loopback address.

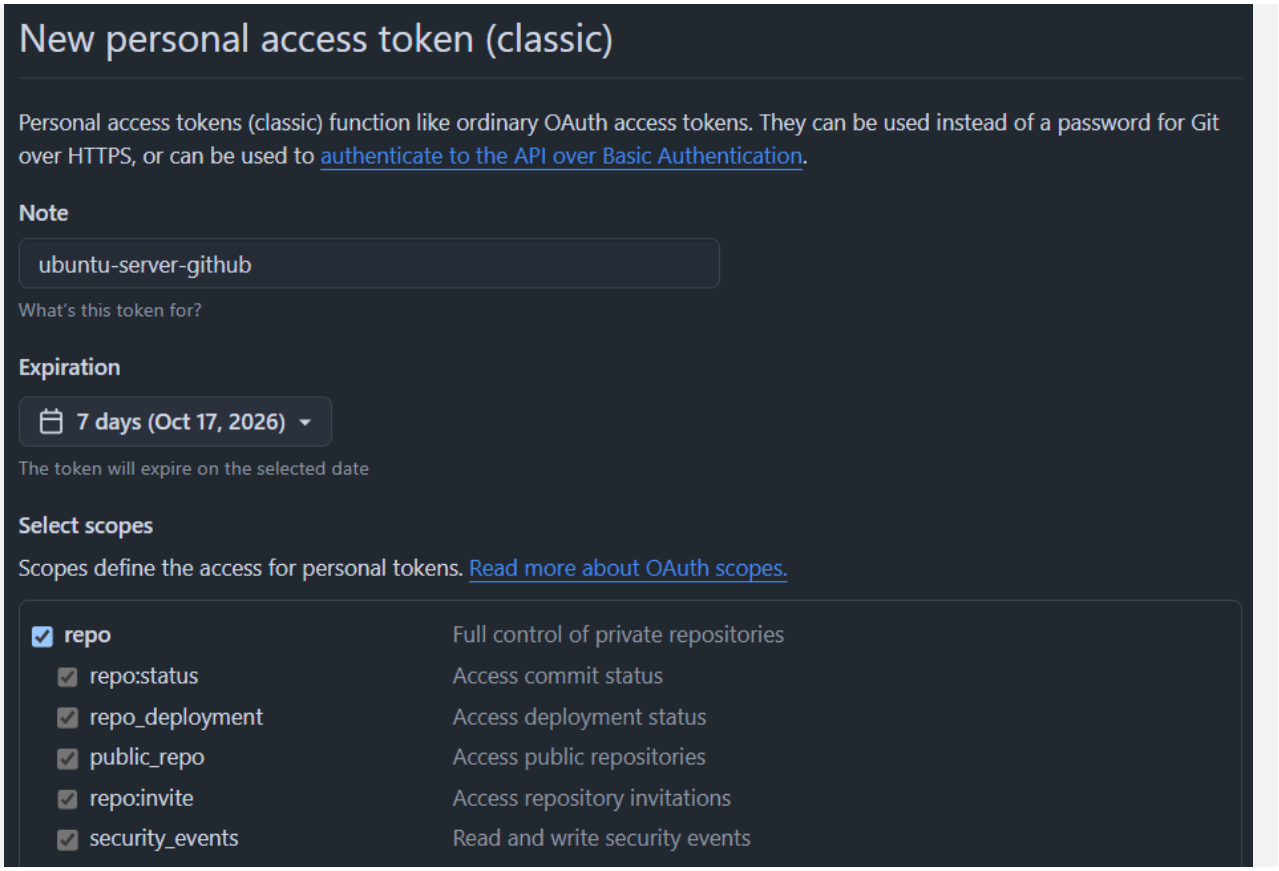
Task 2: Push the Same Repository to GitHub

- **Step 2.1: Create the GitHub Repository and Verify Both Remotes**

An empty public GitHub repository named Assignment01 was created without a README, .gitignore or license.



A GitHub personal access token was generated for authentication. It is not shown in this report.



GitHub was added as the second remote. The output shows both the gitea and github remotes with their fetch and push URLs, and neither URL contains a token or password.

```
hadiaemaanmalik@ubuntu:~/Assignment01$ git remote add github https://github.com/hadiaemaanmalik/Assignment01.git
hadiaemaanmalik@ubuntu:~/Assignment01$ git remote -v
gitea   http://hadiaemaanmalik@127.0.0.1:3000/hadiaemaanmalik/Assignment01.git (fetch)
gitea   http://hadiaemaanmalik@127.0.0.1:3000/hadiaemaanmalik/Assignment01.git (push)
github  https://github.com/hadiaemaanmalik/Assignment01.git (fetch)
github  https://github.com/hadiaemaanmalik/Assignment01.git (push)
hadiaemaanmalik@ubuntu:~/Assignment01$
```

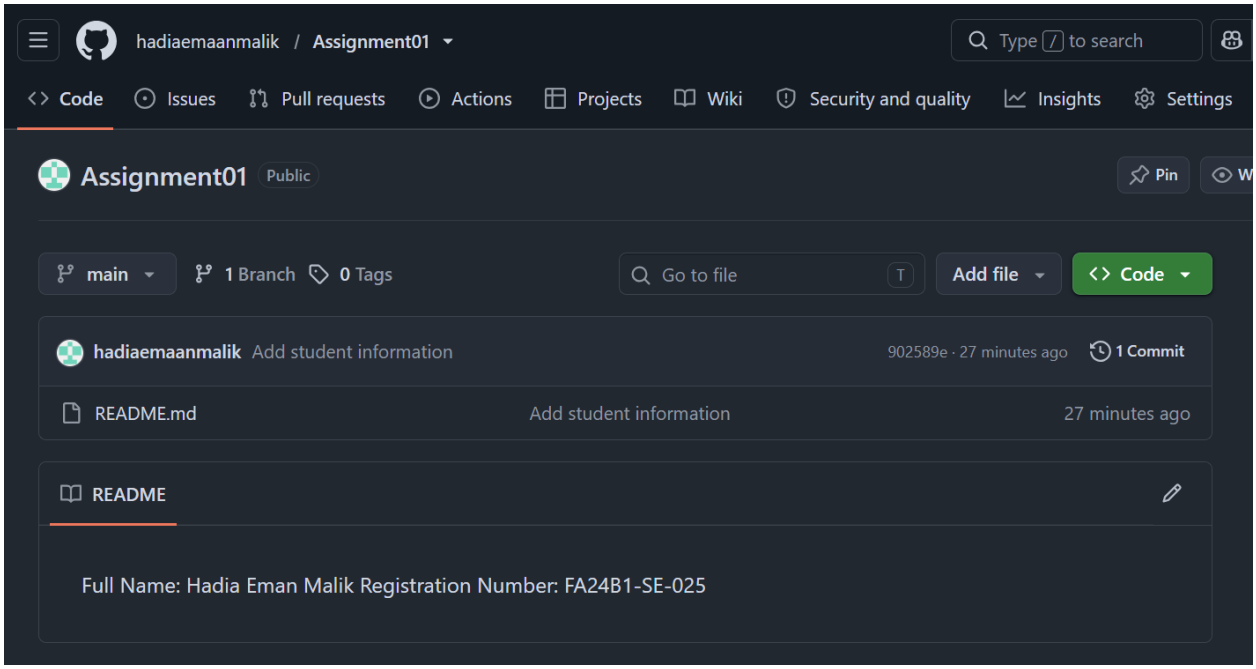
- **Step 2.2: Push to GitHub**

The main branch was pushed to the GitHub remote successfully.

```
hadiaemaanmalik@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': hadiaemaanmalik
Password for 'https://hadiaemaanmalik@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 293 bytes | 146.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hadiaemaanmalik/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
hadiaemaanmalik@ubuntu:~/Assignment01$
```

- **Step 2.3: Verify the GitHub Repository**

The GitHub repository is owned by my account, is named Assignment01, is public, and its rendered README.md shows my full name and registration number.



Task 2 result: The same local repository is connected to both Gitea and GitHub and is publicly available on GitHub.

Task 3: Track Three Large Files with Git LFS

- **Step 3.1: Configure Git LFS**

Git LFS was initialized successfully, its version was confirmed, and tracking was configured for .bin files. The resulting *.bin rule is shown in .gitattributes.

```
hadiaemaanmalik@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
hadiaemaanmalik@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
hadiaemaanmalik@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
hadiaemaanmalik@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
hadiaemaanmalik@ubuntu:~/Assignment01$
```

- **Step 3.2: Create and Verify Three LFS Files**

Three different files, each larger than 100 MiB, were created and staged together with .gitattributes. The command output lists all three files as tracked by Git LFS with their sizes.

```
hadiaemaanmalik@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
hadiaemaanmalik@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
hadiaemaanmalik@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
hadiaemaanmalik@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
hadiaemaanmalik@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
hadiaemaanmalik@ubuntu:~/Assignment01$
```

- **Step 3.3: Commit and Push the LFS Files**

The files were committed and pushed to the GitHub Assignment01 repository. The Git LFS upload completed successfully at 100% (3/3).

```
hadiaemaanmalik@ubuntu:~/Assignment01$ git commit -m "Add three files using Git LFS"
[main ba2f5f9] Add three files using Git LFS
 4 files changed, 10 insertions(+)
 create mode 100644 .gitattributes
 create mode 100644 large-file-1.bin
 create mode 100644 large-file-2.bin
 create mode 100644 large-file-3.bin
hadiaemaanmalik@ubuntu:~/Assignment01$ git push github main
Uploading LFS objects: 100% (3/3), 321 MB | 814 KB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 804 bytes | 134.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hadiaemaanmalik/Assignment01.git
   902589e..ba2f5f9  main -> main
hadiaemaanmalik@ubuntu:~/Assignment01$
```

Task 3 result: Three files larger than 100 MiB are tracked with Git LFS and pushed to the GitHub Assignment01 repository.

Task 4: Create a Portfolio or CV with GitHub Pages

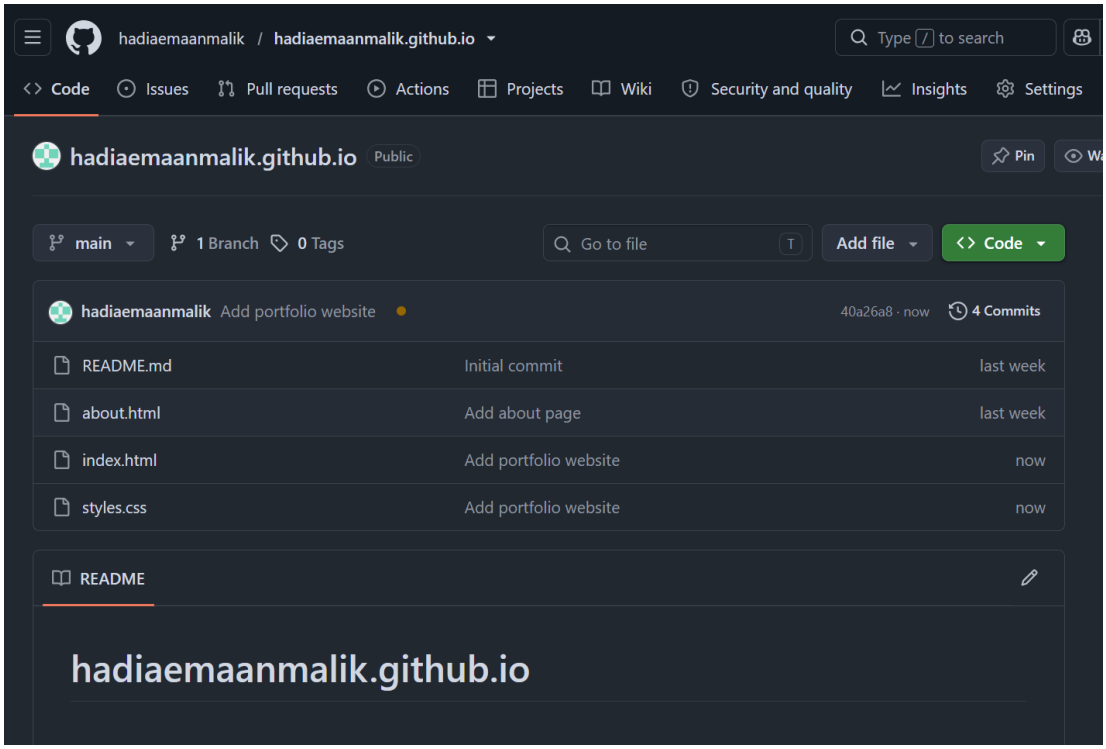
- **Step 4.1: Create and Push the Portfolio Repository**

A public repository named hadiaemaanmalik.github.io was used for the portfolio. The portfolio was written in HTML and CSS using index.html and styles.css. It visibly contains the About Me, Education, Skills and Projects sections. The files were committed and pushed to the main branch.

index.html

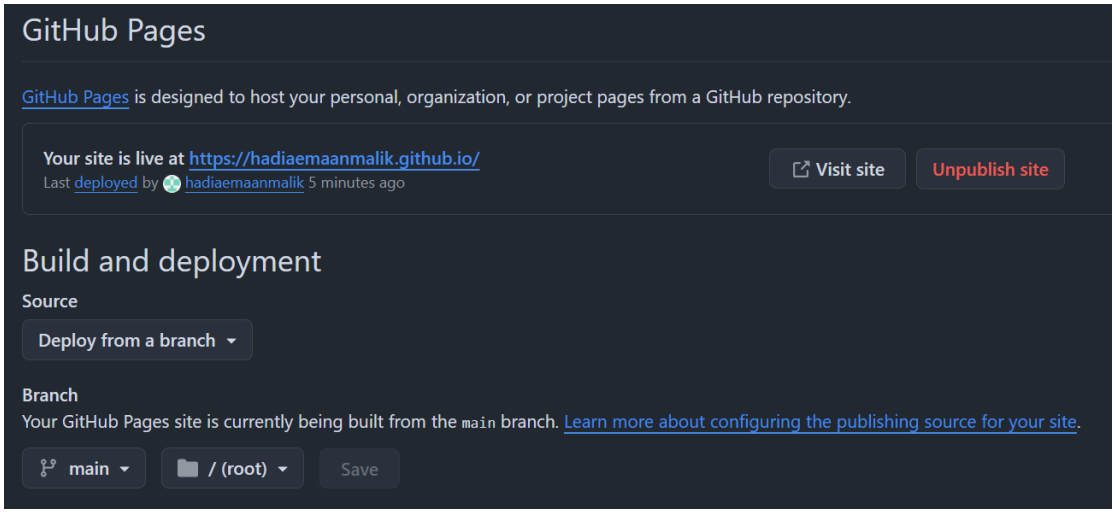
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hadia Eman Malik | Portfolio</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>Hadia Eman Malik</h1>
    <p>BS Software Engineering Student</p>
    <nav>
      <a href="#about">About Me</a>
      <a href="#education">Education</a>
      <a href="#skills">Skills</a>
      <a href="#projects">Projects</a>
    </nav>
  </header>
  <main>
    <section id="about">
      <h2>About Me</h2>
      <p>I am a Software Engineering student at Fatima Jinnah Women University. I enjoy learning cloud computing, version control and building practical projects.</p>
    </section>
    <section id="education">
      <h2>Education</h2>
      <ul>
        <li><strong>BS Software Engineering</strong> - Fatima Jinnah Women University (in progress)</li>
      </ul>
    </section>
    <section id="skills">
      <h2>Skills</h2>
      <ul>
        <li>Git and GitHub</li>
        <li>Linux and Docker basics</li>
        <li>HTML and CSS</li>
      </ul>
    </section>
    <section id="projects">
      <h2>Projects</h2>
      <ul>
        <li><strong>Self-hosted Gitea:</strong> Ran Gitea with Docker on an Ubuntu server.</li>
        <li><strong>Git LFS Practice:</strong> Tracked and pushed large files using Git LFS.</li>
        <li><strong>This Portfolio:</strong> A static site published with GitHub Pages.</li>
      </ul>
    </section>
  </main>
  <footer><p>&copy; 2026 Hadia Eman Malik</p></footer>
</body>
</html>
```

```
# styles.css
# styles.css > footer
1   * { box-sizing: border-box; }
2   body { margin: 0; font-family: Arial, sans-serif; background: #f4f6fb; color: #222; }
3   header { background: #2f6f4f; color: #fff; text-align: center; padding: 30px 10px 0; }
4   header h1 { margin: 0; }
5   nav { background: #245a40; margin-top: 20px; padding: 12px; }
6   nav a { color: #fff; text-decoration: none; margin: 0 14px; font-weight: bold; }
7   main { max-width: 800px; margin: 20px auto; padding: 0 15px; }
8   section { background: #fff; border-radius: 8px; padding: 15px 25px; margin-bottom: 18px; }
9   h2 { color: #2f6f4f; border-bottom: 2px solid #2f6f4f; padding-bottom: 6px; }
10  footer { text-align: center; color: #666; padding: 20px; }
```



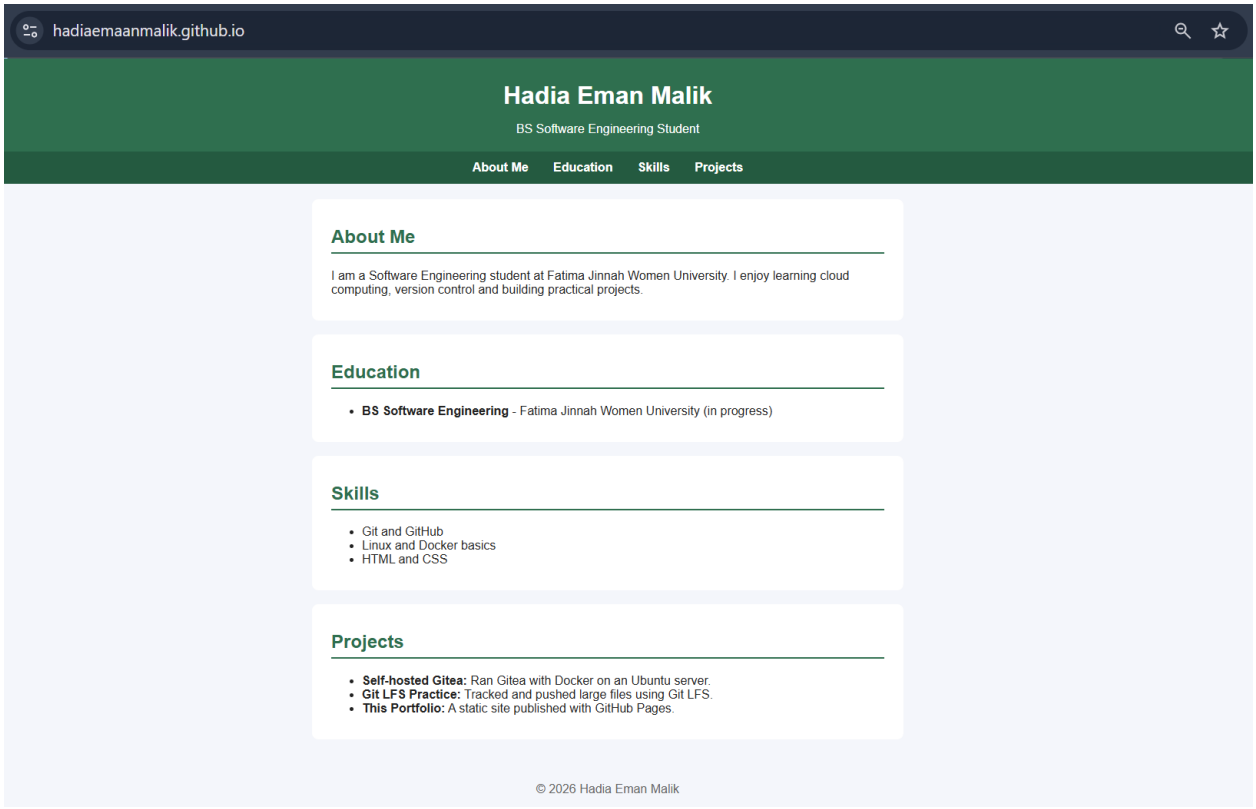
- **Step 4.2: Publish with GitHub Pages**

In the repository Settings, under Pages, Deploy from a branch was selected with the main branch and the / (root) folder. GitHub reports that the site is live at https://hadiaemaanmalik.github.io/.



- **Step 4.3: Verify the Live Portfolio**

The portfolio loads successfully at https://hadiaemaanmalik.github.io/ and shows the About Me, Education, Skills and Projects sections.



Task 4 result: The portfolio is stored in a public GitHub repository and published as a live GitHub Pages site.

Assignment Summary

In this assignment, I installed and ran Gitea on an Ubuntu server using Docker, pushed a local repository to both Gitea and GitHub using two remotes, tracked and pushed three large files with Git LFS, and published a portfolio with GitHub Pages. Authentication was done with personal access tokens entered at the password prompt, so no credentials were stored in remote URLs or shown in screenshots.